

Machine-Checked Status of the SG–Cole–Hopf Route to 3D Navier–Stokes Global Regularity: H1 Obstruction, Stochastic-Lagrangian Repair, and Pancake Closure

The MeTTapedia formalization effort

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Abstract

Ben Goertzel’s manuscript *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program* (v7) reduces global regularity of the 3D incompressible Navier–Stokes equation on $\mathbb{T}^3$ to a uniform-in-time identity-energy bound on the volume-preserving diffeomorphism group $\text{SG} = \text{SDiff}(\mathbb{T}^3)$. The June 2026 MeTTapedia formalization found that the route’s original local adjoint hypothesis (H1) is false in a normalized Fourier-mode shear model: inside every positive H^m chart radius there are area-preserving shears of fixed H^m displacement size whose adjoint $H^m \rightarrow H^m$ ratio is $\sqrt{1 + (\varepsilon k)^2}$ and therefore unbounded. That refutation remains intact.

This July revision records the follow-up repair campaign in full. The corrected route does not rely on the false $H^m \rightarrow H^m$ adjoint bound; it replaces it with a stochastic-Lagrangian/cotangent transport skeleton and a microlocal vorticity-stretching gate. The machine-checked surface now comprises: the evasion contrast showing the H1-breaking shear family leaves pointwise Jacobians untouched; the one-form transport calculus with the exact frame-Laplacian identity; the divergence-free non-exact witness that forces the gauge; the stochastic-Lagrangian pushdown deriving Navier–Stokes with a fully explicit pressure; the conditional BKM chain with its gate consumers; nonzero exact inhabitants and end-to-end canaries; the frozen-strain pancake obstruction with its exact peak-amplification law; the plane-wave Biot–Savart null structure claimed in the pancake-closure proposal; and the dyadic absorption step that would turn the pancake-sector estimate into the BKM bound. The latest audit also isolates why the proposed pancake closure does not yet prove the Millennium statement: its high-low split covers all aperture angles iff the scale-local strain satisfies $\gamma \leq \nu \lambda_q$, exactly the quantitative budget it set out to prove. The surviving open input is the pointwise-in-time Lean predicate `ScaleLocalPancakeStrainBudget`, a sharpened version of the earlier `StochasticStretchingEstimate`. Thus the status is neither a completed Navier–Stokes proof nor a vague “hard analytic gap”: it is a conditional route with one named, scale-local, machine-audited analytic pin.

1 The problem and the role of hypothesis (H1)

1.1 The SG–Cole–Hopf route

We work on the flat three-torus $\mathbb{T}^3$ with smooth divergence-free initial data u_0 . Ben Goertzel’s manuscript *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program*, version v7 (hereafter “v7”), lifts the incompressible Navier–Stokes equation to a diffusion on the volume-preserving diffeomorphism group

$$\text{SG} := \text{SDiff}(\mathbb{T}^3),$$

equipped with a right-invariant Fourier frame $\{D_i\}_{i \geq 1}$ induced by a divergence-free frame $\{E_i\}_{i \geq 1}$ on $\mathbb{T}^3$. The generator and carré du champ are

$$L := \sum_{i \geq 1} D_i^2, \quad \Gamma(F) := \sum_{i \geq 1} (D_i F)^2.$$

One solves the SG heat equation $\partial_t \Phi = \nu L \Phi$ with a strictly positive datum $\varphi = \exp(-W_0/(2\nu))$, applies the Cole–Hopf transform $W = -2\nu \log \Phi$ to obtain an SG Hamilton–Jacobi–Bellman potential, and pushes the Γ -gradient down at the identity to a projected Navier–Stokes solution $u(t)$ on $\mathbb{T}^3$.

The analytic content is concentrated entirely in closing the Beale–Kato–Majda (BKM) vorticity criterion. With $u(t) = \text{Ev}(W(t))$ and $\omega(t) = \text{curl } u(t)$, the push-down identity and Cauchy–Schwarz give a frame constant $C_{\text{curl}E} < \infty$ with

$$\|\omega(t)\|_{L^\infty(\mathbb{T}^3)} \leq \sqrt{C_{\text{curl}E}} \sqrt{\Gamma(W(t))(\text{id})}, \quad (1)$$

so that a uniform-in-time bound on $\Gamma(W(t))(\text{id})$ makes the BKM integral $\int_0^T \|\omega(t)\|_{L^\infty} dt$ finite on every finite interval. By the log chain rule $\Gamma(\log \Phi) = \Gamma(\Phi)/\Phi^2$ and the maximum principle $\Phi(t, \text{id}) \geq m_\varphi$ (v7, Prop. 7.1),

$$\Gamma(W(t))(\text{id}) = 4\nu^2 \Gamma(\log \Phi(t))(\text{id}) \leq \frac{4\nu^2}{m_\varphi^2} \Gamma(P_t \varphi)(\text{id}), \quad (2)$$

which reduces the whole program to a single uniform bound on $\Gamma(P_t \varphi)(\text{id})$.

1.2 Reduction to a local energy bound, and hypothesis (H1)

Rather than invoking a global Bakry–Émery Γ_2 curvature lower bound, v7 obtains that single bound from a *local* package. Let $\kappa : U \rightarrow V \subset H_\sigma^m(\mathbb{T}^3)$ be the chart of a small SDiff^m neighborhood of the identity ($m \geq 5$ integer, with the Γ -Hilbert structure at id equivalent to the H_σ^m norm), let χ be the cutoff defining W_0 and φ , and set the cutoff region

$$K := \kappa^{-1}(\text{supp } \chi) \subset U.$$

Outside K one has $W_0 \equiv 0$, hence $\varphi \equiv 1$ and $\Gamma(\varphi) \equiv 0$. On K , v7 assumes two local hypotheses (v7, §5.3 and §6, displayed at Lemma D.18):

$$(H1) \text{ Adjoint bounded on support: } \exists C_{\text{Ad}} < \infty : \|\text{Ad}_g\|_{H^m \rightarrow H^m} \leq C_{\text{Ad}} \quad \text{for all } g \in K, \quad (3)$$

$$(H2) \text{ Finite cutoff energy: } \exists C_\varphi < \infty : \Gamma(\varphi)(g) \leq C_\varphi^2 \quad \text{for all } g \in K. \quad (4)$$

From (3)–(4) v7 derives the key estimate via a conjugation differentiation identity (v7, Prop. 6.5, proved as Lemma F.7): for all $t \geq 0$,

$$\Gamma(P_t \varphi)(\text{id}) \leq C_{\text{Ad}}^2 C_\varphi^2. \quad (5)$$

The mechanism is transparent. In the flow representation $P_t F(g) = \mathbb{E}[F(G_t \circ g)]$, differentiating along a conjugation curve $C_{t,i}(\varepsilon) = g \circ \exp(\varepsilon E_i) \circ g^{-1}$ produces the direction $\text{Ad}_g E_i$, and Cauchy–Schwarz in the Γ -Hilbert structure gives the pointwise bound

$$|D_{\text{Ad}_g E_i} \varphi(g)|^2 \leq \|\text{Ad}_g\|_{\text{op}}^2 \Gamma(\varphi)(g) \leq C_{\text{Ad}}^2 C_\varphi^2 \mathbf{1}_{\{g \in K\}}, \quad (6)$$

after which a Jensen/energy reduction yields (5). Chaining (5) through (2) and (1) closes BKM. Hypothesis (H1) is therefore the single load-bearing analytic input of the route: *every* subsequent estimate inherits the constant C_{Ad} from (3), and the program establishes global regularity if and only if (3) holds.

1.3 How v7 attempts to verify (H1)

v7 verifies (H1) at Lemma 6.1 (and again at Lemma D.18). The adjoint / push-forward action is written, for $g \in \text{SDiff}^m$ and $v \in H_\sigma^m$, as

$$\text{Ad}_g v := (Dg) (v \circ g^{-1}), \quad (7)$$

and the argument is that, for $m > 5/2$, composition and inversion are continuous (indeed smooth) operations on SDiff^m , so the map $(g, v) \mapsto \text{Ad}_g v$ is continuous from $\text{SDiff}^m \times H_\sigma^m$ into H_σ^m ; hence $g \mapsto \|\text{Ad}_g\|_{\text{op}; H^m \rightarrow H^m}$ is locally bounded near id, and one shrinks K until (3) holds.

The gap is in (7) itself. The factor Dg is the Jacobian of g , and a map $g \in H^m$ has Jacobian $Dg \in H^{m-1}$ only. The adjoint thus carries a *derivative loss*: Ad_g maps H^m into H^{m-1} , and there is no uniform $H^m \rightarrow H^m$ operator bound near the identity in the H^m topology. This is the classical Ebin–Marsden phenomenon: right translation on SDiff^m is smooth, but the adjoint of the right-invariant frame is not a bounded H^m operator uniformly over an H^m -neighborhood. The local-boundedness argument conflates topological-group continuity of composition with the operator-norm estimate (3) that the route actually needs.

2 The Fourier-mode shear model and the checked $\neg(\text{H1})$ theorems

We make the derivative-loss obstruction precise and machine-checkable in a clean analytic model, on the normalized two-torus, of an area-preserving shear family

$$g_k(x, y) = (x, y + a_k \sin(kx)), \quad \delta_k := g_k - \text{id} = (0, a_k \sin(kx)).$$

The amplitude a_k is chosen so that the displacement has *fixed* H^m size ε for every wavenumber k ; thus the whole family sits inside any prescribed chart radius. Applying the push-forward (7) to the constant horizontal field $v = (1, 0)$ gives

$$\text{Ad}_{g_k} v = (1, a_k k \cos(kx)),$$

and, with the same H^m normalization, the operator ratio is

$$\frac{\|\text{Ad}_{g_k} v\|_{H^m}}{\|v\|_{H^m}} = \sqrt{1 + (\varepsilon k)^2}.$$

The displacement stays at size ε while the ratio grows without bound: no finite C_{Ad} can dominate it on any positive chart radius.

This is formalized, with no `sorry` and by elementary means, in the Lean module `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`. The model quantities are defined as follows.

```

/-- In the normalized two-torus shear family, every selected displacement has
fixed 'H^m' size 'epsilon'. -/
def benH1ModeDisplacementHNorm (epsilon : Real) (_k : Nat) : Real :=
  epsilon

/-- Lower-bound ratio obtained by applying the shear push-forward to the
constant horizontal vector field. -/
def benH1ModeAdjointRatio (epsilon : Real) (k : Nat) : Real :=
  Real.sqrt (1 + (epsilon * (k : Real)) ^ 2)

```

```

/-- The linearized commutator part alone grows like 'epsilon * k'. -/
def benH1ModeCommutatorRatio (epsilon : Real) (k : Nat) : Real :=
  epsilon * (k : Real)

```

Hypothesis (H1), restricted to this normalized family inside a chart radius ρ (taking $\varepsilon = \rho/2$ so the family is strictly inside the ball), is the predicate:

```

/-- H1 restricted to the normalized Fourier-mode shear family inside chart
radius 'rho'. -/
def benH1FourierModeUniformBoundWithinRadius (rho C : Real) : Prop :=
  forall k : Nat,
    benH1ModeDisplacementHNorm (rho / 2) k <= rho ->
      benH1ModeAdjointRatio (rho / 2) k <= C

```

The core fact is that the adjoint ratio dominates the linear commutator part εk , which is unbounded over k . The proof is fully elementary: it picks $k > C/\varepsilon$, lower-bounds $\sqrt{1 + (\varepsilon k)^2}$ by εk , and derives a contradiction.

```

theorem benH1ModeAdjointRatio_unbounded {epsilon C : Real}
  (hepsilon : 0 < epsilon) :
  Not (forall k : Nat, benH1ModeAdjointRatio epsilon k <= C) := by
  intro hbound
  obtain ⟨k, hk⟩ := exists_nat_gt (C / epsilon)
  have hC_lt : C < epsilon * (k : Real) := by
    have hmul := mul_lt_mul_of_pos_left hk hepsilon
    field_simp [ne_of_gt hepsilon] at hmul
    exact hmul
  have hnonneg : 0 <= epsilon * (k : Real) := by positivity
  have hlinear_le_ratio :
    epsilon * (k : Real) <= benH1ModeAdjointRatio epsilon k := by
    unfold benH1ModeAdjointRatio
  have hsquare :
    (epsilon * (k : Real)) ^ 2 <= 1 + (epsilon * (k : Real)) ^ 2 := by
    linarith
  calc
    epsilon * (k : Real)
      = Real.sqrt ((epsilon * (k : Real)) ^ 2) := by
        rw [Real.sqrt_sq_eq_abs, abs_of_nonneg hnonneg]
    _ <= Real.sqrt (1 + (epsilon * (k : Real)) ^ 2) :=
      Real.sqrt_le_sqrt hsquare
  exact not_lt_of_ge (le_trans hlinear_le_ratio (hbound k)) hC_lt

```

From this the refutation of (H1) on the model follows directly. Any positive chart radius ρ admits the normalized family at $\varepsilon = \rho/2$ (whose displacements all have H^m size $\rho/2 \leq \rho$), and its adjoint ratios are unbounded, contradicting any proposed finite C .

```

/-- Any positive chart radius contains the normalized shear family with
'epsilon = rho / 2', and its adjoint ratios are unbounded. -/
theorem not_benH1FourierModeUniformBoundWithinRadius {rho C : Real}
  (hrho : 0 < rho) :
  Not (benH1FourierModeUniformBoundWithinRadius rho C) := by
  intro hbound
  have hepsilon : 0 < rho / 2 := by positivity

```

```

have hmode_bound : forall k : Nat, benH1ModeAdjointRatio (rho / 2) k <= C := by
  intro k
  apply hbound k
  unfold benH1ModeDisplacementHNorm
  linarith
exact benH1ModeAdjointRatio_unbounded hepsilon hmode_bound

/-- Compact public verdict: Ben's H1, as a uniform 'H^m -> H^m' adjoint bound,
is false on the normalized Fourier-mode shear model for every positive chart
radius and every proposed finite constant. -/
theorem benGoertzelH1_false_in_fourierModeShearModel :
  forall rho C : Real,
    0 < rho -> Not (benH1FourierModeUniformBoundWithinRadius rho C) := by
  intro rho C hrho
  exact not_benH1FourierModeUniformBoundWithinRadius (rho := rho) (C := C) hrho

```

The three checked statements are `benH1ModeAdjointRatio_unbounded`, `not_benH1FourierModeUniformBoundWithinRadius`, and the compact verdict `benGoertzelH1_false_in_fourierModeShearModel`: for every positive chart radius and every proposed finite constant, the modeled (H1) uniform $H^m \rightarrow H^m$ adjoint bound fails. (A companion lemma `benH1ModeCommutatorRatio_unbounded` records that the linearized commutator part εk is already unbounded on its own.)

3 The numeric lab: the norms are derived, not assumed

The Lean development encodes the two normalized quantities $\|\delta_k\|_{H^m} = \varepsilon$ and $\|\text{Ad}_{g_k} v\|_{H^m} / \|v\|_{H^m} = \sqrt{1 + (\varepsilon k)^2}$ as definitions. To confirm that these are not circular — that the factor k in the adjoint ratio is a genuine consequence of the Sobolev-weight structure and the derivative in (7), rather than an assumption — a companion numeric lab (`ns_ben_h1_fourier_mode_lab.py`, in the MeTTapedia validation lab) *computes* the norms from first quantities. With H^m Fourier weight $w_k = (1 + k^2)^{m/2}$:

- the amplitude is set to $a_k = \sqrt{2} \varepsilon / w_k$, so the displacement norm is $\|\delta_k\|_{H^m} = |a_k| w_k / \sqrt{2} = \varepsilon$ (the $\sqrt{2}$ is the L^2 normalization of $\sin(kx)$);
- differentiating, $\partial_x(a_k \sin(kx)) = a_k k \cos(kx)$, whose H^m norm is $|a_k| k w_k / \sqrt{2} = \varepsilon k$. The weight w_k *cancels* the amplitude, leaving ε ; the surviving factor k is the one derivative gained by Dg_k ;
- the constant field has $\|v\|_{H^m} = 1$, so the adjoint norm is $\sqrt{1^2 + (\varepsilon k)^2}$ and the ratio is $\sqrt{1 + (\varepsilon k)^2}$.

This is exactly the Ebin–Marsden derivative loss of (7) made quantitative: $g_k \in H^m$ has $Dg_k \in H^{m-1}$, and the action of Dg_k multiplies the mode's high-frequency content by its wavenumber. Table 1 lists the lab output for $\varepsilon = 0.01$, $m = 3$. The displacement norm stays fixed at ε across four decades of k while the adjoint ratio grows.

k	amplitude a_k	$\ \delta_k\ _{H^m}$	commutator εk	ratio $\sqrt{1 + (\varepsilon k)^2}$
8	2.699×10^{-5}	0.0100	0.080	1.0032
16	3.433×10^{-6}	0.0100	0.160	1.0127
32	4.310×10^{-7}	0.0100	0.320	1.0500
64	5.393×10^{-8}	0.0100	0.640	1.1873
128	6.743×10^{-9}	0.0100	1.280	1.6243
256	8.429×10^{-10}	0.0100	2.560	2.7484
512	1.054×10^{-10}	0.0100	5.120	5.2167
1024	1.317×10^{-11}	0.0100	10.240	10.2887

Table 1: Fourier-mode lab output ($\varepsilon = 0.01$, $m = 3$). The displacement H^m norm is pinned at ε for every wavenumber, while the adjoint operator ratio $\sqrt{1 + (\varepsilon k)^2}$ grows without bound. For any proposed constant C the ratio first exceeds it near $k \approx \sqrt{C^2 - 1}/\varepsilon$ (e.g. $C = 2$ at $k = 174$, $C = 10$ at $k = 995$, $C = 100$ at $k = 10000$): no finite C_{Ad} survives.

4 The repaired stochastic-Lagrangian route: the full formalized layer

After the H1 obstruction was isolated, the formalization effort tested a different repair strategy. The repaired route abandons the false $H^m \rightarrow H^m$ adjoint estimate and moves the load-bearing estimate to the stochastic-Lagrangian derivative flow of Constantin–Iyer (*Comm. Pure Appl. Math.* **61** (2008), 330–345): the state is a momentum one-form $\mu = u^\flat + dW$ transported along a noisy flow, the velocity is recovered by averaging and projection, and the continuation estimate is a vorticity-stretching gate feeding the BKM criterion. In the Lean development the gate appears first as the coarse input `StochasticStretchingEstimate`, and in the dyadic refinement as the scale-local predicate `ScaleLocalPancakeStrainBudget`.

This section records the *whole* machine-checked surface of the repair campaign, layer by layer, not only the current crux. All Lean listings below are structurally verbatim from the sources (declaration names, hypotheses, and conclusions exactly as checked), with Unicode notation transliterated to ASCII for typesetting; the Lean sources are normative. Table 2 summarizes the layers.

4.1 Why a pointwise transport route can evade the H1 countermodel

The first question for any repair is whether the Section 2 countermodel already defeats it. It does not, and this is itself a theorem. On *the same* H^m -normalized shear family that refutes (H1), the module `ShearJacobianBounds` proves (`shearFamily_pointwise_bounded_while_sobolev_ratio_unbounded`):

1. every pointwise Jacobian deviation is bounded by ε uniformly in the frequency k (`shearFamily_jacobian_uniform_bound`: $\|Dg_k(x)v - v\| \leq \varepsilon\|v\|$ for all x, v); while
2. the $H^m \rightarrow H^m$ operator ratio $\sqrt{1 + (\varepsilon k)^2}$ of the same family admits no finite uniform bound (Section 2).

The high-frequency-shear derivative-loss mechanism therefore lives entirely in the Sobolev operator topology: the quantity feeding a pointwise gate $\int_0^T \mathbb{E}\|DX_t\|_\infty dt$ is not stressed at all by this family. The checked statement is deliberately labeled: it does not prove any pointwise gate — it

Layer (module)	Headline results	Scope label
H1 model (BenH1Break)	adjoint ratio unbounded; $\neg(\text{H1})$	model refutation
Evasion (ShearJacobianBounds)	same family: pointwise Jacobian $\leq \varepsilon$	model contrast
One-forms (OneFormFrameCalculus)	frame-Laplacian identity $\sum_i L_{E_i}^2 = \Delta$	exact identity
Exactness (OneFormExactness)	div-free smooth non-exact witness	exact witness
Gradients (GradientOneForm)	Cartan formula; self-transport identity	exact identities
Pushdown (ConstantinIyerRepresentation)	NS with explicit pressure from transport	conditional (interface)
Instances (DecayingShearInstance, regression)	nonzero exact inhabitant; rest canaries	exact instances
Gate (VorticityStretchingGate)	conditional BKM route, one named pin	conditional
Countermodel (FrozenStrainModel)	exact pancake solution; no uniform depletion	model refutation
Aperture (ApertureCoverageDichotomy)	dichotomy $\iff$ strain budget; gap failures	model equivalence
Null structure (PlaneWaveSelfStrain)	single-mode Biot–Savart null stretching	exact identity
Dyadic closure (DyadicPancakeClosure)	absorption algebra; sharpened pin; route	conditional

Table 2: The formalized surface of the repair campaign (all modules under `Mettapedia.FluidDynamics.NavierStokes`, July 2026). “Model” statements are proved in named analytic models; “exact” statements are unconditional identities or witnesses on $\mathbb{R}^3$ -coordinate space; “conditional” statements are machine-checked implications whose analytic inputs are named hypotheses.

proves this family cannot refute one. That is what licenses moving the route from adjoint bounds to Lagrangian transport.

4.2 The one-form layer: non-exact momenta and the frame Laplacian

The repaired state is a momentum one-form with a gauge, $\mu = u^\flat + dW$, not a scalar potential. Two checked facts pin down why this shape is forced and why the frame generator acts correctly on it.

Non-exact momenta are necessary. Exact one-forms are closed (`oneFormExtDerivAt_eq_zero_of_exact`), and there is a smooth, everywhere divergence-free field that is not exact — the transverse sine mode:

```
theorem exists_divergenceFree_smooth_not_exact_dim3 :
  exists u0 : EuclideanSpace Real (Fin 3) -> EuclideanSpace Real (Fin 3),
    ContDiff Real 2 u0 /\
    (forall x, coordDivergence u0 x = 0) /\
    Not (IsExactOneForm u0)
```

A scalar-potential (pure Cole–Hopf) ansatz therefore cannot carry a general divergence-free momentum: vorticity lives exactly in the non-exact part. This is the constructive face of the Section 1 diagnosis, and it is what the additive gauge dW is for.

The frame reproduces the Laplacian. The route’s generator is a sum of squared frame derivatives ($v7$ ’s $L = \sum_i D_i^2$). On the truncated trigonometric coordinate frame (three axes, transverse polarizations, both phases) the squared Lie derivatives of a one-form sum *exactly* to the coordinate Laplacian: with amplitude $1/\sqrt{2}$ in dimension 3,

$$\sum_j \sum_{p \neq j} \left(L_{\cos\text{-mode}(j,p)}^2 + L_{\sin\text{-mode}(j,p)}^2 \right) \mu = \Delta \mu$$

(`frameLaplacian_oneForm_normalized`, from the general $(\text{card } \iota - 1)a^2$ identity `frameLaplacian_oneForm` and the phase-pair cancellation `lieOneForm_sq_cos_add_sin`).

The supporting calculus (`GradientOneForm`) includes the coordinate Cartan formula for exact forms (`lieOneForm_gradField`: the Lie derivative of dV along u is exact with potential $\langle u, \nabla V \rangle$, i.e. $L_u dV = d\langle u, \nabla V \rangle$ pointwise) and the self-transport identity (`lieOneForm_self`: $L_u u^\flat = (u \cdot \nabla)u + \frac{1}{2}\nabla|u|^2$). These two identities are what turn transport equations into Navier–Stokes with a pressure, next.

4.3 The pushdown theorem: Navier–Stokes with an explicit pressure

The structure `TransportedMomentumData` packages a velocity u , a gauge W , their certified time derivatives, the gradient/Laplacian exchange facts, and one transport hypothesis — the Itô-averaged frame-transport equation for the momentum one-form:

$$\partial_t u + \nabla \dot{W} + L_u(u^\flat + dW) = \nu \Sigma_{\text{frame}}(u^\flat + dW),$$

where Σ_{frame} is the normalized frame second-order sum of Section 4.2 (the hypothesis is stated in averaged form; no stochastic calculus is formalized, and the field is labeled accordingly in the source). From this single hypothesis the pushdown theorem derives the Navier–Stokes equations with a fully explicit pressure:

```
/-- Pushdown theorem: the averaged frame-transport equation for the
momentum one-form u^flat + dW forces u to satisfy the Navier-Stokes
equations with the explicit pressure
p = dW/dt + (1/2)<u,u> + <u, grad W> - nu * Lap W. -/
theorem navierStokes_of_transportedMomentum (t : Real) (x : R3) :
  deriv (fun s => D.u s x) t + fderiv Real (D.u t) x (D.u t x) +
    gradField (D.pressure t) x =
  D.nu smul coordLaplacian (D.u t) x
```

with

$$p = \partial_t W + \frac{1}{2}\langle u, u \rangle + \langle u, \nabla W \rangle - \nu \Delta W.$$

The pressure is not posited: it is assembled from the exact terms of the transport (gauge drift, kinetic energy via the self-transport identity, gauge advection via the Cartan formula, viscous gauge term), and the theorem is unconditional given the data. This is the repaired route’s replacement for `v7`’s identity-energy pushdown: the same endpoint (a genuine Navier–Stokes solution with pressure), reached without the false adjoint bound.

4.4 The vorticity-stretching gate and the conditional route

Continuation is fed by the averaged Cauchy formula $\omega(t) = \mathbb{E}[(DX_t)(\omega_0 \circ A_t)]$, which bounds $\|\omega(t)\|_\infty$ by moments of the flow Jacobian. The gate is the named open input of the coarse route, exactly:

```
/-- The vorticity-stretching gate (named analytic input; OPEN). -/
def StochasticStretchingEstimate
  (D : StochasticCauchyVorticityData Omega) (T : Real) : Prop :=
  IntervalIntegrable D.expectedJacobianBound volume 0 T
```

Its consumers are mechanical and checked: a time-uniform Jacobian bound gives a time-uniform vorticity bound (`vorticity_uniform_bound_of_jacobian_bound`), and under the gate the BKM integrand $t \mapsto \|\omega(t, x)\|$ is interval-integrable with $\int_0^T \|\omega\| \leq \|\omega_0\|_\infty \int_0^T \mathbb{E}\|DX_t\|_\infty$ (`bkm_integrand_intervalIntegrable`, `bkm_integral_bound`). The assembled conditional

route (`stochasticLagrangian_conditional_route`) states: given transported momentum data, averaged-Cauchy vorticity data linked by $\omega = \text{curl} u$, and the gate on $[0, T]$, the velocity satisfies Navier–Stokes with the explicit pressure *and* the BKM integrand is controlled. Every step is machine-checked; the only analytic input is the gate.

The route is inhabited. Non-vacuity is itself checked, twice. The rest state instantiates the route end-to-end (`conditional_route_rest`, with the gate discharged by a constant envelope). Beyond the rest state, the module `DecayingShearInstance` provides a nonzero exact solution of the full transported-momentum system: the decaying shear $u(t, y) = e^{-t} \sin(y_1) e_0$ with gauge amplitude $b(t) = (e^{-2t} - e^{-4t})/8$ on the frequency-doubled mode, whose balance $b' = -4b + \frac{1}{4}e^{-2t}$ (`gaugeAmp_balance`) absorbs the kinetic self-transport term; `decayingShear_navierStokes` confirms the pair satisfies Navier–Stokes with the representation pressure at $\nu = 1$. So the conditional machinery is not vacuously true: its hypotheses have nonzero exact witnesses.

4.5 The frozen-strain countermodel and the aperture geometry

Any proof of the gate must use more than local strain geometry plus stochastic averaging. The frozen-strain model makes this a theorem. For the prescribed strain $A = \text{diag}(2\gamma, -\gamma, -\gamma)$ and the pancake packet (vorticity along the expanding axis, wave vector along a compressive axis), the module `FrozenStrainModel` proves that

$$\omega(t, x) = a(t) \sin(k(t) x_1) e_0, \quad k(t) = k_0 e^{\gamma t}, \quad a(t) = a_0 \exp\left(2\gamma t - \nu k_0^2 \frac{e^{2\gamma t} - 1}{2\gamma}\right)$$

is an *exact* solution of the frozen vorticity equation $\partial_t \omega + (Ax \cdot \nabla) \omega = A\omega + \nu \Delta \omega$ (`frozenStrainVorticity_solves`; the drift, stretching, phase and viscous terms are all checked, not modeled). Writing $g = 2\gamma/(\nu k_0^2)$ for the normalized strain, the amplification peaks at $t^* = \log g/(2\gamma)$ with the exact value

$$\frac{a(t^*)}{a_0} = g e^{1/g-1} \quad (\text{frozenStrainAmplification_peak}),$$

which is unbounded over the strain family: no uniform depletion bound of the form “pancake alignment always acquires diffusive decay at rate $\nu \lambda^2$ ” survives (`frozenStrain_no_uniform_amplification_bound`). Any correct gate proof must therefore use the *self-consistency* of the Navier–Stokes strain — a statement quantified over arbitrary imposed strains is false.

The angular version is the aperture-coverage geometry. Anisotropic Bernstein estimates control cone apertures up to $\lambda^{-1/2}$; stretch–diffusion covariance controls apertures above $\sqrt{\gamma/\nu}/\lambda$. The band claimed by neither,

$$(\lambda^{-1/2}, \sqrt{\gamma/\nu}/\lambda), \quad \text{is nonempty} \iff \nu \lambda < \gamma \quad (\text{anisotropic_coverage_gap}).$$

4.6 July additions: the aperture dichotomy is the strain budget

The July campaign sharpened the coverage gap into an equivalence and closed the remaining escape routes. In `ApertureCoverageDichotomy.lean`:

```
theorem pancakeApertureCovered_iff_strain_le
  (hnu : 0 < nu) (hlam : 0 < lam) :
  (forall theta : Real,
    theta <= (Real.sqrt lam)^{-1} \/\ Real.sqrt (gamma / nu) / lam <= theta)
  <-> gamma <= nu * lam
```

That is: the “narrow $\Rightarrow$ enstrophy-controlled, else $\Rightarrow$ angularly coercive” dichotomy tiles the whole aperture range *exactly when* the scale-local strain budget $\gamma \leq \nu \lambda_q$ holds. On the gap band both mechanisms fail quantitatively: `gapAperture_diffusionRate_lt_strain` proves $\nu(\theta\lambda)^2 < \gamma$ below the stretch–diffusion threshold (the diffusive rate is strictly beaten by the strain), and `gapAperture_bernstein_volume_exceeds_planar_budget` proves $\lambda^2 < \theta^2 \lambda^3$ above the Bernstein threshold (the cone’s Fourier-support volume exceeds the planar budget behind the improved $L^2 \rightarrow L^\infty$ constant).

The time-averaged escape route is closed by a Grönwall-saturation computation: over the coherent window the packet consumes an integrated strain budget of exactly $\int_0^{t^*} 2\gamma dt = \log g$ (`frozenStrain_integratedStrain_peakWindow`) while amplifying to $g e^{1/g-1} \geq e^{\log g-1}$, i.e. amplification at least $e^{\text{budget}-1}$ (`frozenStrainAmplification_peak_ge_exp_integratedStrain_sub_one`). An averaged strain hypothesis therefore bounds amplification only exponentially in itself and cannot substitute for the pointwise-in-time aperture condition in a dyadic relaxation estimate.

4.7 The dyadic pancake closure: verified pieces and the isolated pin

The July repair proposal claims a complete closure of the gate: decompose the weighted stretching sum by angular sector, prove the pancake-sector estimate

$$\int_0^T \sum_{q \geq Q} \lambda_q^{-2} \|A_q \omega_q^{\text{pan}}\|_\infty dt \leq C(T, \nu, u_0, Q) + \varepsilon \int_0^T D_Q(t) dt, \quad D_Q(t) := \sum_{q \geq Q} \|\Delta_q \omega(t)\|_\infty,$$

with $A_q = S(S_{q-L}u)$ the low-frequency strain and ω_q^{pan} the projection onto the pancake cone of aperture $\delta_q = \lambda_q^{-1/2}$, then absorb $\varepsilon \int D_Q$ and conclude $\int_0^T D_Q < \infty$, hence BKM. The formalization adjudicates this claim piece by piece.

The genuinely new ingredient is verified. The proposal’s Step 1 — a pancake vorticity mode does not generate the strain that stretches itself — is checked exactly, not at symbol level. In `PlaneWaveSelfStrain`: for $\omega(x) = \alpha \sin(\kappa x_1) e_0$ with its Biot–Savart velocity (the shear $u(x) = -(\alpha/\kappa) \cos(\kappa x_1) e_2$), `curlField_planeWaveVelocityMode` proves $\text{curl } u = \omega$ exactly, and `strainRateApply_planeWaveVelocityMode_vorticity` proves $S(u)\omega = \frac{1}{2}(\nabla u + (\nabla u)^\top)\omega = 0$ pointwise, with a non-vacuity witness (`planeWaveVorticityMode_ne_zero`). The scope is honest: the identity is for one exact mode; a pancake *sector* aggregates a cone of modes, and charging the cross-mode strain is precisely the open budget below.

The absorption skeleton is checked with explicit constants. The structure `DyadicVorticityStretchingData` carries the four time functionals (D_Q ; pancake, angularly coercive, and residual stretching sums), their nonnegativity and integrability, and the transported dyadic relaxation inequality. The three sector estimates are explicit predicates, e.g.

```
def PancakeSectorEstimate (C eps : Real) : Prop :=
  (Integral of D.pancakeStretchSum over 0..D.T) <=
    C + eps * (Integral of D.highBlockSum over 0..D.T)
```

(coercive and residual analogous; both are *interfaces* here — established in the source analysis by good-unknown/stretch–diffusion and Coifman–Meyer arguments respectively, consumed as named hypotheses, not re-proved). The absorption theorem `highBlockIntegral_le_of_sectorEstimates` proves: sector estimates with total gain $\varepsilon_p + \varepsilon_c + \varepsilon_r < 1$ imply

$$\int_0^T D_Q(t) dt \leq \frac{C_{\text{relax}} + C_p + C_c + C_r}{1 - (\varepsilon_p + \varepsilon_c + \varepsilon_r)},$$

with the a-priori finiteness carried by the integrability fields (supplied in the analytic setting by local-in-time regularity and continuation, which is standard and not re-derived). The theorem `pancakeSectorEstimate_of_scaleLocalStrainBudget` derives the pancake-sector estimate from the pointwise scale-local budget by time integration.

Where the claimed proof fails. The proposal’s high-low split charges the whole non-pancake complement to the angularly coercive estimate. By Section 4.6 that charge is *equivalent* to the scale-local budget $\gamma \leq \nu\lambda_q$ — which is exactly the bound the proposal’s own conditional section had identified as “the remaining burden” before the claimed proof. The claimed proof therefore consumes, at its high-low split, the statement it set out to prove; the closure is circular at precisely that point, and nothing else in it repairs the circle.

The remaining pin is a precise predicate, structurally verbatim:

```

/-- The scale-local pancake strain budget (named analytic input; OPEN --
the first load-bearing missing lemma of the claimed pancake closure). -/
def ScaleLocalPancakeStrainBudget (D : DyadicVorticityStretchingData)
  (B eps : Real) : Prop :=
  exists budget : Real -> Real,
    IntervalIntegrable budget volume 0 D.T /\
    (Integral of budget over 0..D.T) <= B /\
    forall t in Set.Icc 0 D.T,
      D.pancakeStretchSum t <= budget t + eps * D.highBlockSum t

```

Here `pancakeStretchSum` represents $\sum_{q \geq Q} \lambda_q^{-2} \|A_q \omega_q^{\text{pan}}(t)\|_\infty$. The obligation is to establish this pointwise envelope for the *self-consistent* Navier–Stokes strain (generated from the vorticity by the Biot–Savart law), not for an externally imposed frozen strain — via a genuinely Navier–Stokes-specific mechanism (pressure/Hessian feedback, multi-scale Biot–Savart self-depletion, or frequency-cascade accounting, in the proposal’s own taxonomy). The verified plane-wave null structure is evidence in the right direction; the aperture equivalence states exactly what quantitative multi-scale estimate is still missing.

The sharpened conditional route. The headline theorem `stochasticLagrangian_conditional_route_of_pancakeBudget` closes the loop: given transported momentum data, dyadic relaxation data whose block sum dominates the high-frequency part of $\|\text{curl } u\|$ (the Littlewood–Paley decomposition interface), the scale-local pancake strain budget, and the coercive/residual interfaces with total gain below 1, the velocity satisfies Navier–Stokes with the explicit pressure and the BKM integrand is interval-integrable with an explicit bound. This reaches the same endpoint as the coarse conditional route with the gate replaced by the sharper geometric pin; the two pins are cross-referenced in the sources and both remain open. An end-to-end rest-state canary (`rest_conditional_route`) checks that the refined hypotheses are simultaneously satisfiable.

5 Status and scope

We state the scope of this finding exactly.

What is established from the original audit. Hypothesis (H1) of v7 — a uniform $H^m \rightarrow H^m$ operator bound (3) on the adjoint / push-forward Ad_g over an H^m chart-neighborhood of the identity — is *false* in the normalized Fourier-mode shear model of Section 2. The refutation is machine-checked (no sorry, elementary) in `Mettapedia.FluidDynamics.NavierStokes.BenH1Break`, and

the underlying norms are independently derived from the Sobolev-weight structure in the companion numeric lab. The model faithfully captures the mechanism that defeats v7’s local-boundedness argument: the adjoint (7) loses one derivative ($g \in H^m \Rightarrow Dg \in H^{m-1}$), so on a fixed-radius H^m chart the operator norm is unbounded along high-wavenumber shears.

What is not established. This is a refutation of (H1) *in a model* — a single analytic Fourier-mode family together with the linearized commutator action — not a from-Mathlib-Sobolev-first-principles theorem about $\text{SDiff}^m(\mathbb{T}^3)$. The model uses the exact push-forward formula (7) on an explicit shear family and the standard Fourier H^m norm; it does not formalize the full diffeomorphism-group functional analysis. The mechanism (derivative loss) is captured faithfully, but the strength of the conclusion is “checked in this model,” not “checked over the entire chart from Mathlib’s Sobolev theory.”

Consequence for v7. Because every quantitative step of the SG–Cole–Hopf program inherits the constant C_{Ad} from (3) — through (5), (2), and (1) — the falsity of (H1) as stated means the route does *not*, as written, establish global regularity. v7 reduces 3D Navier–Stokes global regularity to (H1) (equivalently, to a uniform curvature / Bakry–Émery-type bound), and that reduction target does not hold uniformly in the form required.

What the repair establishes. The stochastic-Lagrangian repair is not refuted by the H1 countermodel, and that evasion is itself a checked theorem (Section 4.1). Machine-checked without `sorry` across the campaign: the one-form calculus with the exact frame-Laplacian identity; the divergence-free non-exact witness forcing the gauge; the pushdown theorem producing Navier–Stokes with the explicit pressure $p = \partial_t W + \frac{1}{2}\langle u, u \rangle + \langle u, \nabla W \rangle - \nu \Delta W$ from the averaged transport hypothesis; the gate consumers and the assembled conditional route; nonzero exact inhabitants (decaying shear with balanced gauge) and end-to-end rest canaries for both the coarse and the refined route; the frozen-strain exact solution with its unbounded-amplification and coverage-gap theorems; the aperture dichotomy–budget equivalence with both gap-band failure theorems and the Grönwall-saturation bound; the single-mode Biot–Savart null structure; and the dyadic absorption algebra with explicit constants. The theorem `stochasticLagrangian_conditional_route_of_pancakeBudget` records that the route closes if the scale-local pancake strain budget and the non-pancake sector estimates are supplied.

Scope of the new pieces. The same honesty discipline as for the H1 model applies. The frozen-strain results are theorems about a linear prescribed-strain model, labeled as such; the aperture theorems are exact inequalities between the model’s threshold parameters; the plane-wave null structure is an identity for one exact Fourier mode (sector aggregation is exactly the open budget); the averaged transport equation enters the pushdown as a hypothesis field (no stochastic calculus is formalized); and the coercive/residual sector estimates are named interfaces consumed from the source analysis, not re-proved. None of these statements is a claim about general Navier–Stokes solutions beyond what its label says.

Verification metadata. Every headline theorem named in this paper carries a machine axiom audit: each depends on exactly the three standard Lean axioms `propext`, `Classical.choice`, `Quot.sound` — no `sorry`, no additional axioms, no unsafe definitions, and no elaboration-limit overrides anywhere in the campaign’s modules. The Navier–Stokes live-surface and regression

aggregates build green, and every conditional theorem ships with an instantiation canary demonstrating its hypotheses are simultaneously satisfiable.

What remains open. We make no claim to have proved the Millennium Navier–Stokes statement. The latest pancake-closure proposal fails as a complete proof because its aperture split requires the pointwise scale-local condition $\gamma \leq \nu\lambda_q$, and that condition is exactly the burden of the unresolved self-consistent strain estimate. The open analytic input is `ScaleLocalPancakeStrainBudget`, cross-referenced with the coarser `StochasticStretchingEstimate`. A successful completion must prove a pointwise-in-time, scale-local, Biot–Savart self-consistency estimate strong enough to cover the gap band identified above.

Provenance

The manuscript discussed is Ben Goertzel, *Global Regularity of 3D Navier–Stokes via SG–Cole–Hopf Program*, version v7, together with the subsequent pancake-closure repair proposal audited in July 2026. The formalization, Fourier-mode model, repaired conditional route, pancake adjudication, and numeric lab are work of the MeTTapedia formalization effort. The repaired-route development comprises, under `Mettapedia.FluidDynamics.NavierStokes`: `BenH1Break`, `OneFormFrameCalculus`, `OneFormExactness`, `GradientOneForm`, `FrozenStrainModel`, `ApertureCoverageDichotomy`, `PlaneWaveSelfStrain`, and the `StochasticLagrangian` layer (`PullbackOneForm`, `ShearJacobianBounds`, `ConstantinIyerRepresentation`, `VorticityStretchingGate`, `DecayingShearInstance`, `DyadicPancakeClosure`), each with a regression companion. Background references: J. T. Beale, T. Kato, A. Majda, *Comm. Math. Phys.* **94** (1984), 61–66; P. Constantin, G. Iyer, *Comm. Pure Appl. Math.* **61** (2008), 330–345; D. Ebin, J. Marsden, *Ann. of Math.* **92** (1970), 102–163; H. Bahouri, J.-Y. Chemin, R. Danchin, *Fourier Analysis and Nonlinear Partial Differential Equations*, Springer, 2011.